5 Prompt engineering in web applications

This chapter covers

- Prompt engineering to optimize AI model outputs in web applications
- Using few-shot learning to enable quick adaptation of AI models to new tasks
- Using chain-of-thought prompting to improve AI reasoning and problem solving
- Using embeddings for semantic search and content recommendations

Our web application aims to provide engaging AI-powered interactions for users. So far, we've focused on the technical foundations, using cutting-edge tools like Next.js, React, and the Vercel AI SDK to seamlessly integrate various AI models. While this has helped improve the overall user experience, we've relied on the same generic prompt message, instructing the AI to be polite and respond to queries. To offer a truly unique solution, we need to go beyond this one-size-fits-all approach. This is called *prompt engineering*, or the art of carefully crafting and refining the prompts and contextual information submitted to the AI model. By fine-tuning the prompts in this